

STRENGTHS • 'Make in India' concept • Upcoming government policies and plans • Different architectural designs without any compromise on light intensity • Market's focus on quality products rather than quantity	OPPORTUNITIES • Increasing awareness among customers • Increasing market demand • Customer's acceptance to new technology and products
SWOT	
WEAKNESSES • Few manufacturing units • Lack of skilled manpower • Almost no training facilities	THREATS • Availability of LED lighting products from abroad at a competitive price

SWOT analysis



Architectural LED lighting



Panel-integrated solar street-light

troller based solar-charge controller that controls the rate of current drawn from electric batteries. The microcontroller inside the charge controller helps prevent overcharging and against overvoltage that may otherwise hinder the performance of a battery and pose safety risks.

In fact, the microcontroller in the charge controller takes care of a number of tasks, like measuring solar cell voltage and battery voltage, deciding when to start and stop battery charging,

and when to switch on and switch off the load, indicating battery-dip discharge, battery status, charging status and load status with the help of an LED.

Such solar-charge controllers are either stand-alone devices or are integrated within battery packs. These play an important

role in both solar lights and solar LED lights.

Wireless dimmers. Everyone in India dreams of a fully-automated house with an option to control lights. So, what can be better than wirelessly controlling the lights using a mobile phone or tablet?

Today, the market has a new set of dimmers to control the lights. These dimmers are controlled by a home-automation system over Wi-Fi. A lot of free applications are available online to control such dimmers. These are compatible with both constant-current and constant-voltage power supplies, with a flicker-free dimming performance. These dimmers are available for indoor lightings in both industrial as well as residential segments. Some such dimmers have constant-current-limiting and output-voltage-shutdown features to protect lights in faulty situations.

Smart street-light controllers. These controllers have a microcontroller and a real-time clock that does not need to be re-programmed even if the battery fails for a longer duration. It is programmed on the basis of longitude and latitude, as per sunrise and sunset timings adjusted for each month to cater for all seasonal needs.

These have an in-built auto-recovery system for power-failure that facilitates smooth operation of street-lights and a keyboard, along with a relay-control facility, for faster service. The relay has an option to be controlled both manually as well as automatically. It is said to save about

eight per cent of energy against all manual operations, conditional timers and any photo sensors. With the help of such controllers, one can also control street-lights through a computer in one's house. These controllers can control many street-lights at a time.

Passive infra-red (PIR) controllers. PIRs have been in use since quite some time to identify the presence of a human being in a high-security area. But the use of PIRs in lighting systems is totally new. It was not that long ago when I came to know that this type of light-automation is only a concept. In fact, only a year ago, I was discussing with some industry players, how this would soon be possible with the growth of LED lights in the market.

A new PIR controller has a wide range of input voltages that provide flexibility in the configuration of LEDs. It has a wide detection angle of about 102×92 degrees with about 92 detection zones for an effective lighting control. With this type of controller, you do not have to worry about turning off the lights at your home and office when you are away. If the controller detects no human being within its detection range, it would turn off the lights.

These controllers have an adjustable turn-off time, starting from a few seconds to a few hours. Therefore the controller can be adjusted to turn off the lights soon after the last person leaves the covered area.

Filament LEDs. The filament LEDs made in India are designed to light up automotive lights (two-, three- and four-wheelers). Although the patent for these LEDs is still pending in India, these come with life-time warranty. Bulbs made of such LEDs cost the same as incandescent bulbs.

Intensity inspectors. These are portable cameras for measuring the light intensity of LED chip and module, with an effective measuring range of 0.0005 cd/m^2 to 10 cd/m^2 . Designed not only for high-speed measurement for inline inspection, these cameras also measure light intensity of mega-contrast displays.